

Intervals and absolute value equations

Exact determinant ranges of general tridiagonal interval matrices

Difficulty: challenging **Importance:** interesting to specialist

Status: Solved

Topic: interval linear algebra; determinants; computational complexity

Rating rationale: Challenging reflects classifying exact determinant-range computation for a structured but dependency-sensitive interval family; specialist impact concerns this particular interval matrix class.

Independently reviewed resolution - 2026-09-11

Complexity classification. Theorem 1 proves NP-completeness of the upper determinant threshold and NP-hardness of exact determinant-range computation, even for regular independent-entry tridiagonal interval matrices. Section 5 supplies the exact-output upper bound: a polynomial algorithm for the full displayed target exists if and only if $P = NP$. No unconditional separation or strong NP-hardness is asserted.

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. See the [complete manuscript](#), [independent agent review](#) and [submission record](#). The source archive identifies the drafts as AI-generated; authorship is recorded at the submitter's request. This is agent verification, not external human peer review or formal proof-assistant certification.

The difficulty, importance and rating rationale below are historical assessments of the original open target. The original statement and dated audits are preserved.

Problem statement

For $n \geq 2$, let the input consist of $3n - 2$ closed real intervals with rational endpoints: diagonal intervals $[a_i, \bar{a}_i]$ for $1 \leq i \leq n$, upper-diagonal intervals $[b_i, \bar{b}_i]$, and lower-diagonal intervals $[\underline{c}_i, \bar{c}_i]$ for $1 \leq i < n$. All lower endpoints are at most their upper endpoints. Define

$$\mathcal{T} = \{T \in \mathbb{R}^{n \times n} : T_{ii} \in [a_i, \bar{a}_i], \quad T_{i,i+1} \in [b_i, \bar{b}_i], \quad T_{i+1,i} \in [\underline{c}_i, \bar{c}_i], \quad T_{ij} = 0 \text{ if } |i - j| > 1\}.$$

All uncertain entries vary independently. Does a deterministic algorithm exist that returns the two exact rational numbers

$$d_- = \min_{T \in \mathcal{T}} \det T, \quad d_+ = \max_{T \in \mathcal{T}} \det T$$

in time polynomial in the total binary input length? The target is the exact range $[d_-, d_+]$, including instances containing singular matrices and intervals crossing zero. Rational output is appropriate because the determinant is affine in each entry separately and its extrema are attained at endpoint matrices. The polynomial bound must be uniform in n .

References

Horáček, Hladík, and Matějka, *Determinants of interval matrices*, Electronic Journal of Linear Algebra 33 (2018), 99–112, §5.4, p. 106, immediately after Proposition 5.6; [journal PDF](#). The corresponding location in [arXiv:1809.03736v1](#) is §6.4, p. 9, after Proposition 6.7. The source proves polynomial computability for interval tridiagonal H-matrices and explicitly leaves general tridiagonal interval matrices open.

Status check (2026-09-08)

Searches for *tridiagonal interval determinant complexity*, *tridiagonal determinant range*, and *tridiagonal interval determinant polynomial 2026* found no resolution for independent-entry exact ranges. Two superficially conflicting Thirupathi–Thamaraiselvan papers, *Symbolic Algorithm for Inverting General k-Tridiagonal Interval Matrices* and *A Symbolic Algorithm for Solving Doubly Bordered k-Tridiagonal Interval Linear Systems*, both published in 2023, use different generalized arithmetic (§2.2, pp. 3–4). Their multiplication centers the result at the product of midpoints and takes the smaller distance to the standard product endpoints. Consequently it sends $[1, 2]$ and $[2, 3]$ to $[2, 11/2]$, while independent products range over $[2, 6]$. Their symbolic determinant algorithms therefore do not establish exact ranges in the sense defined above.

Audit update — 2026-09-10

Rechecked Hladík’s [primary paper](#), §5.4 after Proposition 5.6. The tridiagonal interval H-matrix subclass has a polynomial exact determinant-range algorithm and is a substantive included part of the displayed target, so the status is Partially resolved. The general tridiagonal complexity question remains unresolved. Later determinant-range searches found no classification in standard independent-entry interval arithmetic; the generalized-arithmetic claims discussed above use a different operation. Impact is narrowed to specialist. The remaining general question has a historical explicit source, without a recent reaffirmation.